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Identification of Optimum Filter Steady-State Gain for Systems with Unknown Noise Covariances

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***Abstract*—A discrete linear stationary system is considered for which the input noise covariance Q and the output noise covariance R are unknown. A stable filter with a suboptimal gain is assumed. An identification scheme is presented which uses the autocorrelation functions of the innovations sequence of the suboptimal filter to determine the optimum filter steady state gain Γ directly without the intermediate determination of the unknown covariances Q and R. The approach used is to identify an output equivalent representation of the original system which does not involve the unknown covariances directly.**

I. Introduction

In the design of a Kalman–Bucy filter, the noise covariances Q and R are assumed known. In many practical situations Q and R are either unknown or known only approximately; in this case, the system performance is at best suboptimum. Several authors [3], [4], [5], [11] have presented schemes for the identification of the unknown covariances with varying degrees of success. A good summary was recently presented by Mehra [15]. The use of the innovations sequence in the identification of the unknown covariances was introduced by Mehra in [3] where he also presents an algorithm to estimate the optimum steady-state filter gain directly. The identification scheme presented here starts with correlation measurements and uses an iterative algorithm which is shown to be a contraction mapping under the mild assumption of complete controllability and complete observability of the given system. Numerical examples are presented, and comparisons are made with Mehra's method as given in [3].

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II. Statement of Problem

Consider a discrete linear stationary multivariable stochastic system

$$x_{i+1} = Fx_i + Gu_i \tag{1a}$$

$$y_i = Hx_i + v_i \tag{1b}$$

$$\dim x_k = n, \dim y_k = r, \dim u_k = p \tag{2a}$$

$$E\{x_0\} = 0, E\{x_0x_0'\} = \Sigma_0 \tag{2b}$$

where $\{u_i\}$ and $\{v_i\}$ are independent stationary white-noise sequences with zero-mean values and covariances Q and R, respectively. It is assumed that the system is completely observable and completely controllable.

The innovations sequence $\{\nu_i\}$ and the estimation error

$\tilde{x}_{i|i-1}$ for the system (1a),(1b) are defined, respectively, by

$$v_i \triangleq y_i - H\hat{x}_{i|i-1} \tag{3}$$

$$\tilde{x}_{i|i-1} \triangleq x_i - \hat{x}_{i|i-1} \tag{4}$$

where $\hat{x}_{i|i-1}$ is the optimum linear estimate of x_i. For an optimum filter in the steady state, the innovations sequence is a stationary white noise sequence with covariance W given by [7]

$$W = H\Sigma H' + R \tag{5}$$

where Σ, the optimum steady-state error covariance, satisfies the equation

$$\Sigma = F\Sigma F' - F\Sigma H'(H\Sigma H' + R)^{-1}H\Sigma F' + GQG'. \tag{6}$$

Using (3) and (4), we have

$$y_i = H\hat{x}_{i|i-1} + v_i \tag{7a}$$

and as is well known [2], in the steady state,

$$\hat{x}_{i+1|i} = F\hat{x}_{i|i-1} + \Gamma v_i \tag{7b}$$

$$\Gamma = F\Sigma H'W^{-1} \tag{7c}$$

where Γ is the steady-state optimum filter gain. A system described by the (7a),(b) is an alternate representation of the output sequence $\{y_i\}$ of the system described by (1a),(b). Thus, for the purpose of filter design, the sequence $\{y_i\}$ may be assumed generated by either of the equations (1a),(b) or (7a),(b). The use of (7a),(b) has obvious advantages [12]; the steady-state Kalman filter for this representation is simply

$$\hat{x}_{i+1|i} = F\hat{x}_{i|i-1} + \Gamma(y_i - H\hat{x}_{i|i-1}) \tag{8}$$

and the corresponding steady-state error covariance is zero.

Given data from a filter (8) with a suboptimum gain Γ_D, it is desired to obtain the optimum filter gain Γ and the covariance W of the innovations sequence of the optimum filter.

III. Estimation of Γ and W

3.1. Autocorrelation Functions

The steady-state suboptimum filter is given by

$$x^*_{i+1|i} = Fx^*_{i|i-1} + \Gamma_D(y_i - Hx^*_{i|i-1}) \tag{9}$$

where $x^*_{i|i-1}$ denotes the suboptimum estimate of $\hat{x}_{i|i-1}$ and Γ_D is the suboptimum gain. The suboptimum filter steady-state error covariance is defined by

$$P^* \triangleq E\{(\hat{x}_{i+1} - x^*_{i+1|i})(\hat{x}_{i+1} - x^*_{i+1|i})'\}. \tag{10}$$

Using (7a),(b), and (9), we obtain

$$P^* = (F - \Gamma_D H)P^*(F - \Gamma_D H)' + (\Gamma_D - \Gamma)W(\Gamma_D - \Gamma)'. \tag{11}$$

Following Mehra [3] the autocorrelation functions of the innovations sequence of the suboptimum filter in the steady state are obtained from

$$\begin{aligned} C_j &= E\{v_i^* v_{i-j}^{*\prime}\} \\ &= E\{(y_i - Hx^*_{i|i-1})(y_{i-j} - Hx^*_{i-j|i-j-1})'\} \\ &= \begin{cases} H(F - \Gamma_D H)^{j-1}[(F - \Gamma_D H)P^*H' - \Gamma_D W + \Gamma W], & j \neq 0 \\ HP^*H' + W, & j = 0. \end{cases} \end{aligned} \tag{12}$$

For the optimum filter, $\Gamma_D = \Gamma$ and $P^* = 0$; hence, from (12), $C_j = 0$ for $j \neq 0$. Define,

$$B' \triangleq [H' \vdots F'H' \vdots \cdots \vdots (F')^{n-1}H']. \tag{13}$$

B, an $nr \times n$ matrix of rank n, is the system observability matrix. From (12) and (13), we obtain

$$B(FP^*H' + \Gamma W) = \begin{bmatrix} C_1 + H\Gamma_D C_0 \\ C_2 + H\Gamma_D C_1 + HF\Gamma_D C_0 \\ \cdot \\ \cdot \\ \cdot \\ C_n + H\Gamma_D C_{n-1} + \cdots + HF^{n-1}\Gamma_D C_0 \end{bmatrix} \triangleq A. \tag{14}$$ [1]

The autocorrelation functions $C_j (j = 0,1,2,\cdots,n)$ can be estimated experimentally. Hence, the $n \times r$ matrix $(FP^*H' + \Gamma W)$ can be solved for from (14). The solution is given by

$$(FP^*H' + \Gamma W) = B^\dagger A \tag{15}$$

where $B^\dagger$ is the generalized inverse of B defined by

$$B^\dagger = (B'B)^{-1}B' \tag{16}$$

which exists, since the system is completely observable.

3.2. Iterative Algorithm to Determine Γ and W

From (11), (12), and (15) we have three simultaneous matrix equations for Γ, W, and P^*.

$$W = C_0 - HP^*H' \tag{17}$$

$$\Gamma = (B^\dagger A - FP^*H')W^{-1} \tag{18}$$

$$P^* = (F - \Gamma_D H)P^*(F - \Gamma_D H)' + (\Gamma_D - \Gamma)W(\Gamma_D - \Gamma)'. \tag{19}$$

Let

$$W(X) = C_0 - HXH' \tag{20}$$

$$\Gamma(X) = (B^\dagger A - FXH')W^{-1}(X) \tag{21}$$

$$T(X) = (F - \Gamma_D H)X(F - \Gamma_D H)' + (\Gamma_D - \Gamma(X))W(X)(\Gamma_D - \Gamma(X))' \tag{22}$$

where $X \in R^{n\times n}$ is positive semidefinite. $T(X)$ defines a mapping of $R^{n\times n}$ into itself. We shall show that it is a contraction and that we can therefore construct a converging algorithm. We define a set $\zeta \subset R^{n\times n}$ as follows:

$$\zeta = \{X : X \text{ positive semidefinite}, X \leq P^* + \Sigma\}. \tag{23}$$

[1] Steps leading to (14) are given in detail in Appendix II.

The proof will take the following steps:

a) Show that $(F - \Gamma(X)H)$ is asymptotically stable if $X \in \zeta$.

b) Use a) to show that ζ is T-invariant.

c) Show that $\|T(X_1) - T(X_2)\| \leq \alpha\|X_1 - X_2\|$, $X_1, X_2 \in \zeta$; $\quad 0 \leq \alpha < 1$

Lemma 1: Let P be an arbitrary $n \times n$ positive semidefinite matrix and R an arbitrary $r \times r$ positive definite matrix. Let F be an $n \times n$ asymptotically stable (in the context of discrete systems) matrix, and H an arbitrary $r \times n$ matrix.

If

$$\Gamma = FPH'(HPH' + R)^{-1}$$

then $F - \Gamma H$ is asymptotically stable.

Proof: Let $K = PH'(HPH' + R)^{-1}$. $\quad (24)$

By the matrix inversion lemma [13],

$$\begin{aligned}(I - KH)^{-1} &= (I - PH'(HPH' + R)^{-1}H)^{-1} \\ &= I + PH'R^{-1}H. \qquad (25)\end{aligned}$$

Since $H'R^{-1}H$ is positive semidefinite, the eigenvalues of $PH'R^{-1}H$ can be shown to be real and nonnegative, and thus the eigenvalues of $I + PH'P^{-1}H$ are real and greater than or equal to 1. This implies that the eigenvalues of $I - KH$ are real, and lie between 0 and 1.

Since $F - \Gamma H = F(I - KH)$ and the spectral radius of a matrix, denoted by $\rho(\cdot)$, is a norm, we have

$$\rho(F - \Gamma H) \leq \rho(F) \cdot \rho(I - KH). \qquad (26)$$

Since F is asymptotically stable, all its eigenvalues are less than 1 in magnitude, and the result follows.

We use Lemma 1 to establish that $\Gamma(X)$ is a stabilizing gain. Q.E.D.

Lemma 2: $(F - \Gamma(X)H)$ is asymptotically stable if $X \in \zeta$.

Proof: From (17) and (18)

$$C_0 = W + HP^*H' \qquad (27)$$

$$B\dagger A = \Gamma W + FP^*H'. \qquad (28)$$

Using (20) and (21),

$$\begin{aligned}\Gamma(X) = (\Gamma W + FP^*H' - FXH')(W + HP^*H' \\ - HXH')^{-1}. \qquad (29)\end{aligned}$$

Using (7c) and (5) for Γ and W,

$$\Gamma(X) = F(\Sigma + P^* - X)H'[H(\Sigma + P^* - X)H' + R]^{-1}. \qquad (30)$$

Clearly, $X \in \zeta$ implies

$$\Sigma + P^* - X \geq 0. \qquad (31)$$

Since $R > 0$, the conditions of Lemma 1 are fulfilled, and $\Gamma(X)$ is a stabilizing gain.

We now establish the fact that $T(\cdot)$ maps ζ into itself. Q.E.D.

Lemma 3: $X \in \zeta$ implies $T(X) \in \zeta$.

Proof: Adding (6) and (11), and using (7c), we obtain after some algebra,

$$\begin{aligned}P^* + \Sigma = (F - \Gamma_D H)(P^* + \Sigma)(F - \Gamma_D H)' \\ + \Gamma_D R \Gamma_D' + GQG'. \qquad (32)\end{aligned}$$

Define $S(X) \triangleq P^* + \Sigma - X$. Then we note that

$$W(X) = HS(X)H' + R \qquad (33)$$

$$\Gamma(X) = FS(X)H'W^{-1}(X). \qquad (34)$$

Using this in (22),

$$\begin{aligned}T(X) &= (F - \Gamma_D H)X(F - \Gamma_D H)' + \Gamma_D R \Gamma_D' \\ &\quad + \Gamma_D HS(X)H'\Gamma_D' - \Gamma_D HS(X)F' \\ &\quad - FS(X)H'\Gamma_D' + FS(X)H'W^{-1}(X)HS(X)F' \\ &= (F - \Gamma_D H)(X + S(X))(F - \Gamma_D H)' + \Gamma_D R \Gamma_D' \\ &\quad - F[S(X) - S(X)H'W^{-1}(X)HS(X)]F'. \qquad (35)\end{aligned}$$

Using (32),

$$\begin{aligned}T(X) &= P^* + \Sigma - GQG' \\ &\quad - F[S(X) - S(X)H'W^{-1}(X)HS(X)]F'. \qquad (36)\end{aligned}$$

Since $S(X) \geq 0$, the last term may be written as

$$\begin{aligned}FS^{\frac{1}{2}}(X)[I - S^{\frac{1}{2}\prime}(X)H'[HS^{\frac{1}{2}}(X)S^{\frac{1}{2}\prime}(X)H' + R]^{-1} \\ \cdot HS^{\frac{1}{2}}(X)]S^{\frac{1}{2}\prime}(X)F'. \qquad (37)\end{aligned}$$

Using the matrix inversion lemma, this becomes

$$FS^{\frac{1}{2}}(X)[I + S^{\frac{1}{2}\prime}(X)H'R^{-1}HS^{\frac{1}{2}}(X)]^{-1}S^{\frac{1}{2}\prime}(X)F'. \qquad (38)$$

Clearly, this is at least positive semidefinite. Since GQG' is also at least positive semidefinite, it follows from (36) that

$$T(X) \leq P^* + \Sigma. \qquad (39)$$

Q.E.D.

Lemma 4: Given $X_1, X_2 \in \zeta$, there exists an $\alpha, 0 \leq \alpha < 1$, such that

$$\|T(X_2) - T(X_1)\| \leq \alpha\|X_2 - X_1\|$$

where the norm is the Euclidian norm,

$$\|X\| = \Big(\sum_{i,j} x_{ij}^2\Big)^{\frac{1}{2}}.$$

Proof: According to one of the mean value theorems [14],

$$\begin{aligned}\|T(X_2) - T(X_1)\| \leq \sup_{0 \leq t \leq 1} \|\mathfrak{L}[T(tX_1) \\ + (1 - t)X_2)]\|_N \|X_2 - X_1\| \qquad (40)\end{aligned}$$

where $\mathfrak{L}$ is the derivative operator and $\|\cdot\|_N$ is the operator norm. It turns out here [14] that the proper operator norm is the spectral norm,

$$\|\mathfrak{L}T(X)\|_N = \rho\left[\left(\frac{d \operatorname{vec} T}{d \operatorname{vec} X}\right)'\left(\frac{d \operatorname{vec} T}{d \operatorname{vec} X}\right)\right]^{\frac{1}{2}}. \qquad (41)$$

Using a basic inequality for matrix norms,

$$\|\mathfrak{L}T(X)\|_N \leq \rho\left[\left(\frac{d \text{ vec } T}{d \text{ vec } X}\right)'\right]^{\frac{1}{2}} \cdot \rho\left(\frac{d \text{ vec } T}{d \text{ vec } X}\right)^{\frac{1}{2}} = \rho\left(\frac{d \text{ vec } T}{d \text{ vec } X}\right). \quad (42)$$

We show in Appendix I that

$$\frac{d \text{ vec } T}{d \text{ vec } X} = (F - \Gamma(X)H)' \otimes (F - \Gamma(X)H)'. \quad (43)$$

Since the eigenvalues of a Kronecker product are the products of the eigenvalues,

$$\rho\left(\frac{d \text{ vec } T}{d \text{ vec } X}\right) = \rho(F - \Gamma(X)H)^2. \quad (44)$$

Since ζ is compact, the right-hand side (RHS) of (44) has a maximum at some member $X_{\max}$ of the set. Since $F - \Gamma(X)H$ is asymptotically stable for all $X \in \zeta$, then

$$\rho(F - \Gamma(X_{\max})H) = \alpha_0 < 1. \quad (45)$$

Thus, for any $X \in \zeta$

$$\|\mathfrak{L}T(X)\|_N \leq \alpha_0{}^2 = \alpha < 1. \quad (46)$$

It is simple to show that ζ is convex, i.e., $[tX_1 + (1 - t)X_2] \in \zeta$ for any $0 \leq t \leq 1$ if $X_1, X_2 \in \zeta$. Thus

$$\|T(X_2) - T(X_1)\| \leq \alpha\|X_2 - X_1\|, \qquad \alpha < 1. \quad (47)$$

Q.E.D.

Finally, we state the desired result.

Theorem 1: Let $X_0 \in \zeta$. Then the sequence $X_0, X_1, X_2, \cdots$ formed by the repeated operation $X_{m+1} = T(X_m)$ converges to P^*.

Proof: By Lemma 3, and since $X_0 \in \zeta$, we have $X_m \in \zeta$ for all m. Furthermore, P^* is obviously in ζ. Thus,

$$\|X_{m+1} - P^*\| = \|T(X_m) - T(P^*)\| \leq \alpha\|X_m - P^*\| \quad (48)$$

and the sequence converges, the error norm being bounded by $\alpha^m\|X_0 - P^*\|$. Q.E.D.

Remark: A simple choice for X_0 is the null matrix. This is further justified by the fact that $P^* = 0$ if Γ_D happens to be equal to Γ.

For convenience, we restate the algorithm below:

$$W(X_m) = C_0 - HX_mH' \quad (49)$$

$$\Gamma(X_m) = (B\dagger A - FX_mH')W^{-1}(X_m) \quad (50)$$

$$X_{m+1} = (F - \Gamma_DH)X_m(F - \Gamma_DH)' + (\Gamma_D - \Gamma(X_m))W(X_m)(\Gamma_D - \Gamma(X_m))'. \quad (51)$$

3.3. The Algorithm of Mehra [3]

The corresponding equations in the algorithm of Mehra [3] are reproduced here for the purpose of comparison.

$$\delta\hat{M}_j = F(I - \hat{K}_jH)\delta\hat{M}_j(I - \hat{K}_jH)'F' - F(\hat{K}_j - \hat{K}_{j-1})\hat{C}o(\hat{K}_j - \hat{K}_{j-1})'F' \quad (52)$$

$$\hat{M}_{j+1}\hat{H}' = \hat{M}_j\hat{H}' + \delta\hat{M}_jH' \quad (53)$$

$$\hat{K}_{j+1} = \hat{M}_{j+1}\hat{H}'(H\hat{M}_{j+1}\hat{H}' + \hat{R})^{-1}. \quad (54)$$

For each iteration j of the algorithm, the Lyapunov-type matrix equation (52) has to be solved. In the algorithm presented here, this is not required; it involves only matrix multiplications.

IV. Numerical Example

The method of this paper is applied to the same system that was used by Mehra [3] from inertial navigation. The system matrices are given by

$$F = \begin{bmatrix} 0.75 & -1.74 & -0.3 & 0.0 & -0.15 \\ 0.09 & 0.91 & -0.0015 & 0.0 & -0.008 \\ 0.0 & 0.0 & 0.95 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.55 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.905 \end{bmatrix},$$

$$G = \begin{bmatrix} 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \\ 24.64 & 0.0 & 0.0 \\ 0.0 & 0.835 & 0.0 \\ 0.0 & 0.0 & 1.83 \end{bmatrix}$$

$$H = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 & 1.0 \\ 0.0 & 1.0 & 0.0 & 1.0 & 0.0 \end{bmatrix}$$

$$Q = \text{diag }(1.0,1.0,1.0)$$
$$R = \text{diag }(1.0,1.0). \quad (55)$$

Procedure: Initially, Q and R are assigned the values Q_D and R_D, respectively. Using Q_D and R_D a Kalman filter is designed from which the suboptimum filter steady-state gain Γ_D is obtained. Now assuming that both the system and the filter have reached the steady state, we proceed as follows:

Step 1) Obtain estimates of the autocorrelation functions of the innovations sequence of the suboptimum filter from

$$C_j = \frac{1}{N}\sum_{i=1}^{N-j} \mathbf{v}^*_{i+j}\mathbf{v}_i{}^{*\prime}, \qquad j = (0,1,2,\cdots,n) \quad (56)$$

where N is the number of observations. The estimates given by (56) are biased for finite N, but asymptotically they are unbiased and consistent.

Step 2) Using Γ_D and C_j ($j = 0,1,2,\cdots,n$) estimates P^*, W, and Γ are obtained iteratively from (49)–(51).

The iterative scheme (49)–(51) converges uniquely to the optimum gain Γ if the autocorrelation functions C_j are accurately known. However, due to the finiteness of the sample size and other experimental errors, the C_j's may not be accurately known and this sets a limit on the accuracy with which Γ can be determined.

It is suggested by Mehra [3] that the estimate of Γ could be improved by repeating Steps 1) and 2) above with the new gain. This is not the case in general. Suppose, for

TABLE I
COMPARISON OF FILTER PERFORMANCE

RUN NO.	A – METHOD OF MEHRA B – PRESENT METHOD	P_{11}	P_{22}	P_{33}	P_{44}	P_{55}	$\|P\|$
1	A	96.4	6.83	1534	4.57	47.8	1556
	B	80.1	2.39	1250	1.80	22.2	1309
2	A	107	9.81	1625	5.60	57.7	1646
	B	79.9	2.03	1247	1.49	20.0	1278
3	A	(Algorithm of Mehra [3] Diverges)					
	B	106	1.66	1256	1.69	56.5	1287
4	A	79.2	1.57	1266	1.40	21.1	1297
	B	74.2	1.26	1224	1.05	13.9	1258
5	A	79.6	2.83	1343	1.99	25.4	1371
	B	75.4	1.46	1238	1.19	16.1	1270
	Optimum Filter	72.3	1.14	1213	0.932	11.7	1245

example, we start with the optimum gain. If there are any errors in the estimation of the autocorrelation functions C_j due to the finiteness of sample size, the algorithm in step 2) will not lead to the optimum gain we started with. Thus, in general, the estimate of Γ can only be improved by improving the estimates of the autocorrelation functions.

In the following example, the autocorrelation functions $C_j(j = 0,1,2,\cdots,n)$ were estimated, in each run, using 1000 observations of the innovations sequence $\{\boldsymbol{v}_i^*\}$. Comparisons are made below of the performances of the filter using the gain obtained by the present method and that using the gain obtained by the method of Mehra [3]. The evaluation of filter performance is based on the steady-state error covariance, using model (1a),(b), that is attained by the filter. When the filter gain is optimum, the corresponding error covariance is minimum.

Example 1: Q_D and R_D are as used in [3].

$$Q_D = \text{diag}\ (0.25,0.5,0.75)$$
$$R_D = \text{diag}\ (0.4,0.6).$$

The performances of the filter obtained by the method of [3] and that of this paper are summarized in Table I. The results show that the present method leads to a filter with smaller error covariance than that of [3]. In run number 3, the algorithm of Mehra [3] did not converge; this indicates the sensitivity of the algorithm to errors in the autocorrelation functions.

It is also observed experimentally that when Q_D and R_D are chosen as $Q_D = \text{diag}\ (q,q,q)$, $R_D = \text{diag}\ (r,r)$ where $q = 1.0$ and $r = 0.05$, the algorithm of Mehra [3] does not converge even when the autocorrelation functions C_j have their true values. Divergence occurred whenever $q > 10r$.

V. CONCLUSION

In this paper, an algorithm is presented for the direct identification of the optimum filter steady-state gain Γ for systems with unknown noise covariances Q and R. The algorithm is shown to be a contraction mapping under the assumption of complete controllability and complete observability of the given system. This guarantees the unique convergence of the algorithm by the contraction mapping principle. The algorithm is shown at least in one example, to be computationally easier and of greater accuracy than that presented by Mehra [3]. In particular, the stability of the algorithm appears to be less dependent on the assumed values of the unknown noise covariance.

APPENDIX I

Let $\langle X\rangle_{ij}$ denote the (i,j) element of X and J_{ij} an $n \times n$ matrix with 1 in the (i,j) position and 0's elsewhere $(dT(X)/d\langle X\rangle)$ is the derivative of the matrix $T(X)$ with respect to the (i,j) element of X. From (22) and taking derivatives, we obtain

$$\begin{aligned}\frac{dT(X)}{d\langle X\rangle_{ij}} &= (F - \Gamma_D H)J_{ij}(F - \Gamma_D H)' \\ &\quad + \frac{d(\Gamma_D - \Gamma(X))}{d\langle X\rangle_{ij}} W(X)(\Gamma_D - \Gamma(X))' \\ &\quad + (\Gamma_D - \Gamma(X))\frac{dW(X)}{d\langle X\rangle_{ij}}(\Gamma_D - \Gamma(X))' \\ &\quad + (\Gamma_D - \Gamma(X))W(X)\cdot\frac{d(\Gamma_D - \Gamma(X))'}{d\langle X\rangle_{ij}}. \qquad \text{(I-1)}\end{aligned}$$

For clarity $\langle X\rangle_{ij}$ and J_{ij} will be noted simply by $\langle X\rangle$ and J, respectively. Using (20) and (21),

$$\begin{aligned}\frac{d(\Gamma_D - \Gamma(X))}{d\langle X\rangle} &= \frac{d}{d\langle X\rangle}\{\Gamma_D - (B\dagger A - FXH') \\ &\qquad \cdot (C_0 - HXH')^{-1}\} \\ &= FJH'W(X)^{-1} - \Gamma(X)HJH'W(X)^{-1} \\ &= (F - \Gamma(X)H)JH'W(X)^{-1} \qquad \text{(I-2)}^2\end{aligned}$$

[2] We acknowledge helpful discussions with R. Tait [8] on matrix derivatives.

$$\frac{dW(X)}{d\langle X\rangle} = -HJH'. \tag{I-3}$$

Substituting (I-2) and (I-3) into (I-1)

$$\begin{aligned}\frac{dT(X)}{d\langle X\rangle} &= (F - \Gamma_D H)J(F - \Gamma_D H)' \\ &\quad + (F - \Gamma(X)H)JH'(\Gamma_D - \Gamma(X))' \\ &\quad - (\Gamma_D - \Gamma(X))HJH'(\Gamma_D - \Gamma(X))' \\ &\quad + (\Gamma_D - \Gamma(X))HJ(F - \Gamma(X)H)'. \end{aligned}\tag{I-4}$$

Using Kronecker products [8],[9], we obtain directly from (I-4)

$$\begin{aligned}\frac{d\,\mathrm{vec}\,T}{d\,\mathrm{vec}\,X} &= (F - \Gamma_D H)' \otimes (F - \Gamma_D H)' + H'(\Gamma_D - \Gamma(X))' \\ &\quad \otimes (F - \Gamma(X)H)' - H'(\Gamma_D - \Gamma(X))' \\ &\quad \otimes H'(\Gamma_D - \Gamma(X))' + (F - \Gamma(X)H)' \\ &\quad \otimes H'(\Gamma_D - \Gamma(X))'. \end{aligned}\tag{I-5}$$

Multiplying out and collecting terms (I-5) becomes

$$\frac{d\,\mathrm{vec}\,T}{d\,\mathrm{vec}\,X} = (F - \Gamma(X)H)' \otimes (F - \Gamma(X)H)'. \tag{I-6}$$

Appendix II

Using (12) we obtain

$$\begin{aligned} C_1 &= H(FPH' + \Gamma W) - H\Gamma_D C_0 \\ C_2 &= HF(FPH + \Gamma W) - H\Gamma_D C_1 - HF\Gamma_D C_0 \\ &\cdot \\ &\cdot \\ &\cdot \\ C_n &= HF^{n-1}(FPH' + \Gamma W) - H\Gamma_D C_{n-1} \\ &\quad - HF\Gamma_D C_{n-2} - \cdots - HF^{n-1}\Gamma_D C_0. \end{aligned}\tag{II-1}$$

(II-1) can be written as

$$\begin{aligned} H(FPH' + \Gamma W) &= C_1 + H\Gamma_D C_0 \\ HF(FPH' + \Gamma W) &= C_2 + H\Gamma_D C_1 + HF\Gamma_D C_0 \end{aligned}\tag{II-2}$$

$$\begin{aligned} &\cdot \qquad\qquad \cdot \\ &\cdot \qquad\qquad \cdot \\ &\cdot \qquad\qquad \cdot \\ HF^{n-1}(FPH' + \Gamma W) &= C_n + H\Gamma_D C_{n-1} + \cdots \\ &\quad + HF^{n-1}\Gamma_D C_0, \end{aligned}\tag{II-3}$$

i.e., $B(FPH' + \Gamma W) = A$.

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